The Empathetic Buddy

MISSION: **Decoding the Implicit in Mental Health Support**

TEAM: **Data AImers**

Text-to-Text Chatbots

DESCRIPTION: Standard RAG systems fail in mental health because they respond to the literal text. If a user deflects with sarcasm or numbness, standard AI provides generic, templated advice that feels robotic and disconnected.

State-to-Text Orchestration

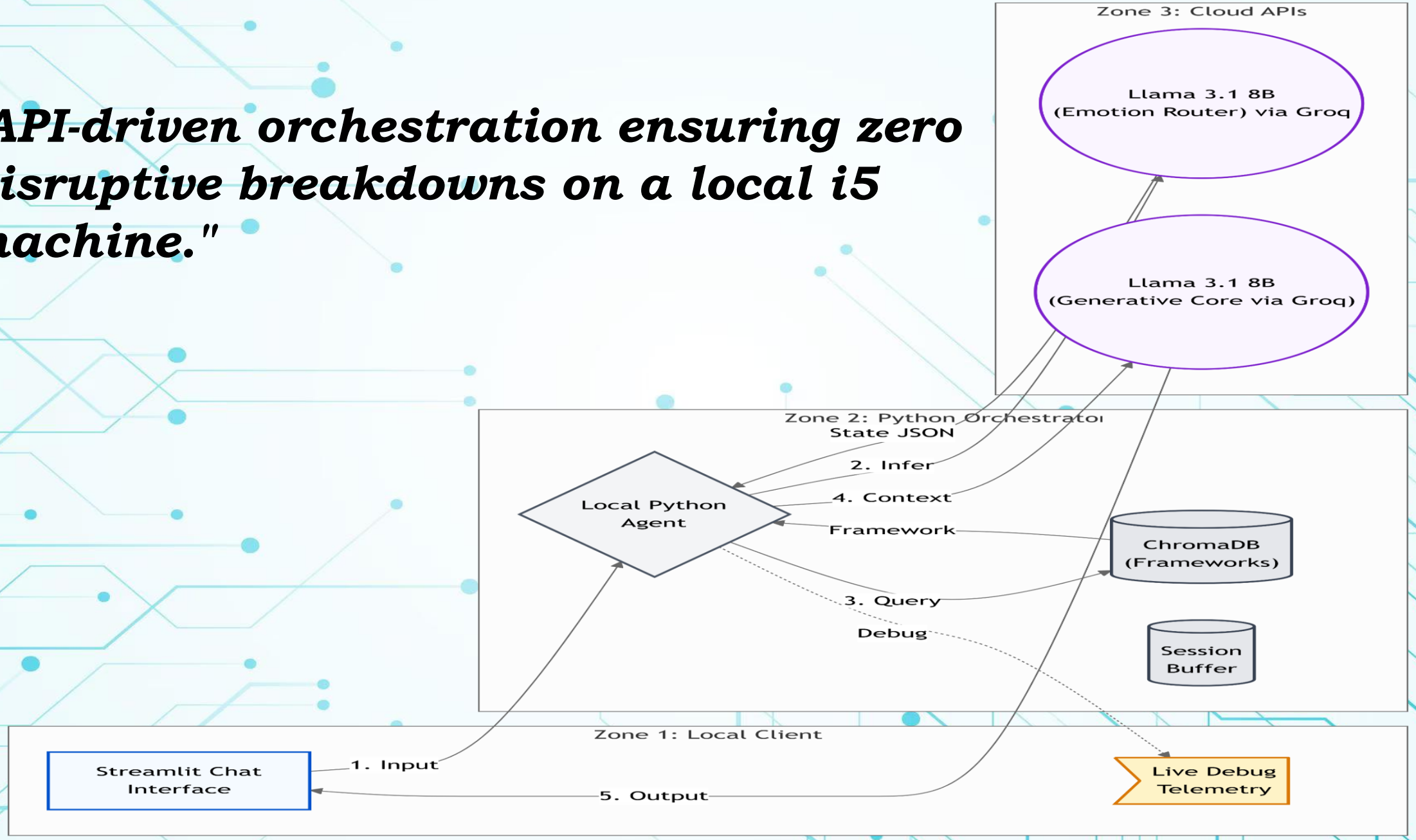
DESCRIPTION: We built a multi-model pipeline that separates understanding from responding.

CORE CAPABILITIES:

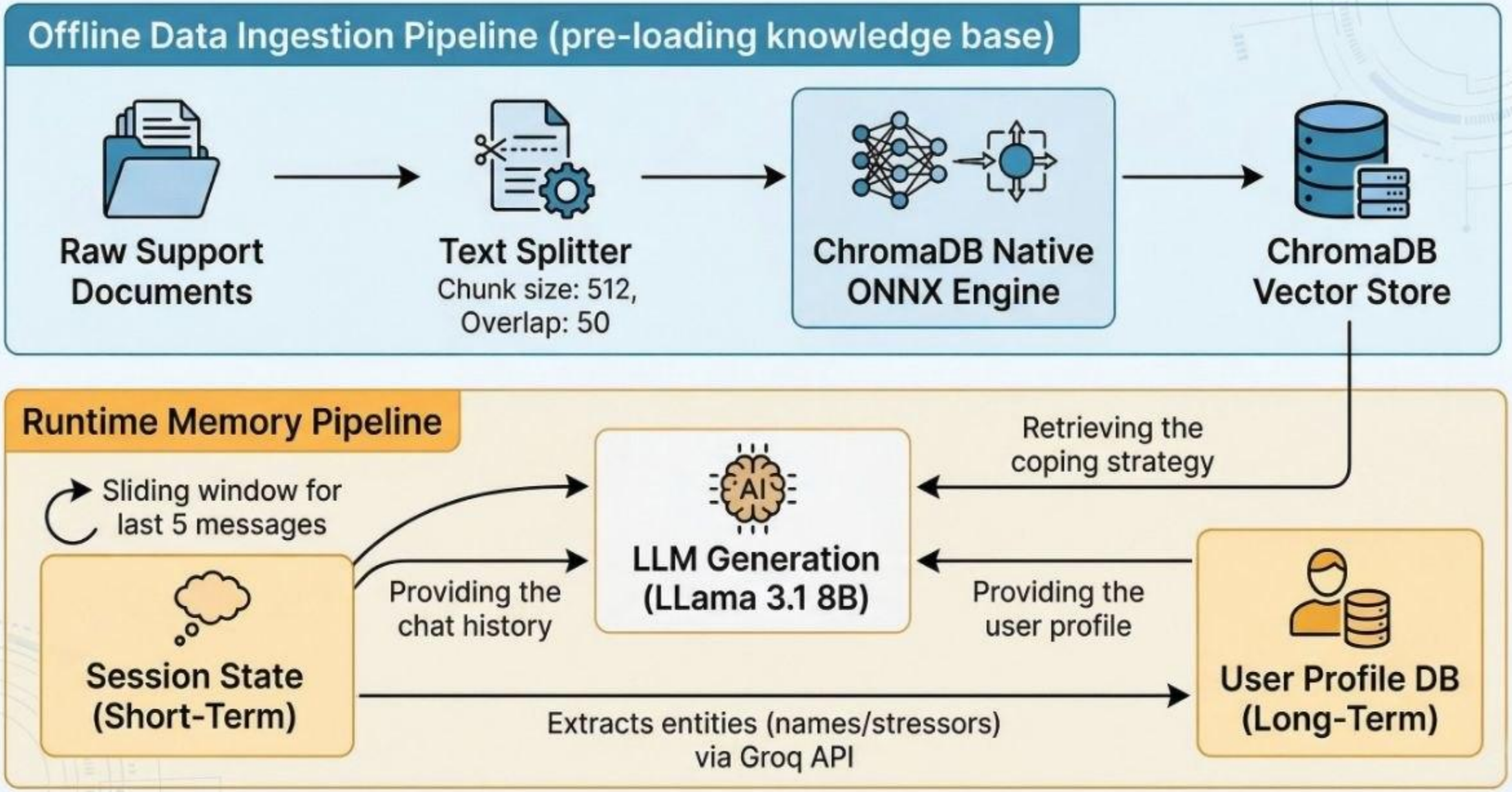
1. ***Implicit Inference:*** Decodes unstated distress (sarcasm/numbness) before drafting a reply.
2. ***Clinical Grounding:*** Retrieves targeted CBT/DBT frameworks via local, zero-latency vector memory.
3. ***Stateful Continuity:*** Tracks named entities and stressors across turns to ensure zero contradictions.

High-level System Architecture

"API-driven orchestration ensuring zero disruptive breakdowns on a local i5 machine."



Rag & Memory Pipeline



Technical Stack

❑ FRONTEND UI

TECHNOLOGY USED: Streamlit

- ✓ **Use:** Provides a fast, lightweight, and stable chat interface for the live demo.
- ✓ **Edge:** Ensures zero UI crashes on the local machine during complex multi-turn judging.

❑ LLM-1

TECHNOLOGY USED: groq LLAMA 3.1 8B
(Emotion Router)

- ✓ **Use**: High-speed psychological analyst.
- ✓ **Edge**: Ultra-low latency and infinite API limits to guarantee 100% uptime during the live demo.

❑ VECTOR DB

TECHNOLOGY USED: ChromaDB
(Local, low-latency)

- ✓ **Use**: Stores and retrieves clinically anchored psychological frameworks (like CBT techniques).
- ✓ **Edge**: Runs locally for ultra-low latency, bypassing the need for slow, unstable web scraping.

❑ FRAMEWORK

TECHNOLOGY USED: PYTHON

- ✓ **Use:** The central nervous system connecting the local UI, local database, and cloud APIs.
- ✓ **Edge:** Keeps the data flow organized and seamlessly updates the user's long-term JSON profile in the background.

❑ LLM-2

TECHNOLOGY USED: groq LLAMA 3.1 8B
(Generative Core)

- ✓ **Use:** Synthesizes the extracted emotion, chat history, and retrieved clinical frameworks.
- ✓ **Edge:** Leverages its massive context window to craft deeply empathetic, personalized responses without losing historical continuity.

❑ EMBEDDINGS

TECHNOLOGY USED: **ChromaDB Default ONNX Engine**
(Fast, dense vector generation)

- ✓ **Use:** Converts text into searchable vectors for the RAG pipeline.
- ✓ **Edge:** Bypasses Windows OS PyTorch dependency blocks by computing vectors natively inside ChromaDB

Edge Case Resolution

PROBLEM	SYSTEM COMPONENT	OUTCOME
Sarcasm/Implicit Distress	Llama 3.1	Extracts true emotional state as JSON
Multi-Turn Entity Resolution	Short-Term Session Buffer	Maps pronouns to historical context
Hallucination Control	Vector Grounding + Regex Safety Check	Prevents generic or dangerous medical advice
Passive vs. Active Risk (e.g., "I just want to sleep" vs. imminent harm)	Local Regex Guardrail	Instantly triggers hardcoded crisis UI overlay, bypassing slow LLM generation
Help-Rejecting Complaining (e.g., "Nothing works")	Llama 3.1 + Pro Prompting	Detects resistance and dynamically shifts generation from "advice" to "pure validation"

"The Empathetic Engine shifts AI from a reactive search engine to a proactive support system—delivering clinically grounded empathy at scale."